

GEOTECHNICAL INVESTIGATION REPORT

Cottonwood Creek Pumping Plant (PP-2)
Clearwater Valley Regional Water Conveyance System

Prepared for:

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1.0 INTRODUCTION

1.1 Project Description

The Clearwater Valley Irrigation District (CVID) is constructing the Clearwater Valley Regional Water Conveyance System, a multi-phase infrastructure program to upgrade water delivery from Cottonwood Creek to agricultural and municipal users throughout the district. Phase 1 includes construction of Cottonwood Creek Pumping Plant No. 2 (PP-2), a reinforced concrete pump station housing three 500-horsepower vertical turbine pumps with a combined design capacity of 15,000 gallons per minute.

The pumping plant will be constructed on the south bank of Cottonwood Creek approximately 2 miles northeast of the town of King Hill in Elmore County, Idaho. The structure will include a below-grade pump bay (approximately 45 feet by 25 feet in plan, extending 12 feet below existing grade), an at-grade motor control center and electrical room, and a reinforced concrete discharge manifold connecting to a 36-inch transmission main. An access road and retaining wall will be constructed along the south perimeter of the site.

1.2 Purpose and Scope

Western Geotechnical Consultants (WGC) was retained by CVID to perform a geotechnical investigation for the PP-2 pumping plant. The purpose of this investigation was to evaluate subsurface conditions at the site and develop geotechnical engineering recommendations for design and construction of the facility. The scope of services included:

- Drilling, logging, and sampling of four exploratory borings to depths of 25 to 40 feet below existing grade;
- Standard penetration testing (SPT) at 2.5-foot intervals through the alluvial overburden;
- HQ wireline diamond coring of the underlying bedrock formation;
- Installation of groundwater monitoring standpipes in each boring;
- Laboratory testing of representative soil and rock samples including classification, strength, swell, consolidation, and corrosion testing; and
- Engineering analysis and development of geotechnical recommendations for foundation design, earthwork, swell mitigation, corrosion protection, and construction considerations.

1.3 Site Description

The project site is located on an alluvial terrace along the south bank of Cottonwood Creek at an elevation of approximately 5,484 feet above mean sea level. The terrace surface is relatively flat with a gentle slope (approximately 2 percent) toward the creek to the north. At the time of our field exploration in September 2018, the site was vegetated with native grasses and sagebrush. Cottonwood Creek was flowing at a moderate level approximately 150 feet north of the proposed pump bay location.

The surrounding area is characterized by gently rolling terrain underlain by Tertiary volcanic and volcanoclastic sedimentary rocks of the Challis Group, mantled by Quaternary alluvial deposits along the creek drainage. Agricultural lands (irrigated pasture and hay fields) adjoin the site to the south and east. An existing CVID diversion structure and canal headworks are located approximately 800 feet upstream (east) of the site.

2.0 FIELD EXPLORATION

2.1 Drilling Program

Four exploratory borings, designated BH-PP2-01 through BH-PP2-04, were drilled at the site between September 10 and September 14, 2018. Drilling was performed by Mountain West Drilling, Inc. of Boise, Idaho using a truck-mounted CME-55 drill rig. The borings were advanced through the alluvial overburden using 4.25-inch inside-diameter hollow-stem augers. Upon encountering bedrock, the borings were advanced by HQ-size (2.50-inch core diameter) wireline diamond coring. Boring locations were selected to characterize subsurface conditions beneath each major structural element of the facility.

Table 2-1: Boring Summary

Boring	Location	Ground Elev. (ft)	Total Depth (ft)	GW Depth (ft)	Core Start (ft)
BH-PP2-01	Pump bay centerline	5,484.7	40.0	13.5	21.0
BH-PP2-02	Discharge manifold (60 ft NE)	5,483.9	35.0	14.2	23.0
BH-PP2-03	Intake structure (80 ft SW)	5,485.2	30.0	13.8	20.5
BH-PP2-04	Access road / retaining wall	5,484.1	25.0	14.0	22.0

2.2 Sampling and In-Situ Testing

Soil samples were obtained from the alluvial overburden at 2.5-foot intervals using a standard 2-inch outside-diameter split-spoon sampler driven by a 140-pound automatic hammer falling 30 inches (ASTM D1586). The number of blows required to drive the sampler the final 12 inches of an 18-inch drive is recorded as the standard penetration resistance (N-value). Relatively undisturbed samples of the clay stratum were obtained using 3-inch diameter Shelby tube samplers (ASTM D1587) at selected depths for consolidation and swell testing.

Rock core was obtained using a double-tube HQ wireline core barrel equipped with a split inner tube. Core was logged, photographed, and placed in wooden core boxes in the field. Representative specimens were selected and sealed in wax for transport to the laboratory for unconfined compressive strength and indirect tensile strength testing.

2.3 Groundwater Observations

Upon completion of drilling, each borehole was fitted with a 1-inch PVC standpipe screened across the water-bearing alluvial sand stratum. Groundwater levels were measured approximately 24 hours after installation of the standpipes. Measured static water levels ranged from 13.5 to 14.2 feet below ground surface (bgs) across the four borings, with an average depth of approximately 13.9 feet bgs. The water table was encountered within the alluvial sand and silt stratum (Layer 3) at all boring locations. Seasonal fluctuations of plus or minus 2 feet are anticipated, with the highest water levels occurring during spring snowmelt and irrigation season (March through June).

3.0 SUBSURFACE CONDITIONS

3.1 Regional Geology

The project site is located within the western Snake River Plain physiographic province, near the transition to the Idaho Batholith to the north. Bedrock in the area consists of Eocene-age volcanoclastic sedimentary rocks and tuffs of the Challis Volcanic Group. These rocks were deposited in a series of intermontane basins and consist of fine-grained sandstone, siltstone, tuffaceous mudstone, and interbedded ash-fall tuffs. The sequence is generally flat-lying to gently dipping (5 to 10 degrees to the south) and moderately indurated.

The bedrock is overlain by Quaternary alluvial deposits associated with the Cottonwood Creek drainage. These deposits consist of floodplain clays and silts, channel sands and gravels, and terrace alluvium. The site is situated on a low alluvial terrace approximately 15 to 20 feet above the present creek level, suggesting the alluvial materials were deposited during late Pleistocene to Holocene time. Geologic mapping by the Idaho Geological Survey (Lewis et al., 2012) designates the site as Qal (Quaternary alluvium) overlying Tcs (Tertiary Challis sedimentary rocks).

3.2 Subsurface Profile

The subsurface conditions encountered in the four borings were generally consistent across the site, with minor variations in layer thickness. The subsurface profile may be generalized as follows:

Layer 1 — Alluvial Fill / Topsoil (0 to 2-4 feet bgs): Brown, loose to medium dense, silty sand with gravel fragments (SM/GM). The upper 12 inches contain a root zone with organic matter. This layer varies from approximately 2 feet thick at BH-PP2-03 (near the creek bank where erosion has thinned the cover) to 4 feet at BH-PP2-01. This material is not suitable for foundation bearing and will be stripped during construction.

Layer 2 — Alluvial Clay (2-4 feet to 10-12 feet bgs): Olive-brown to brown, firm to stiff, lean clay with sand (CL). This stratum is the primary foundation bearing material for shallow footings. The clay is of low to moderate plasticity with liquid limits ranging from 35 to 41 and plasticity indices of 17 to 20. The clay fraction (finer than 0.002 mm) ranges from 22 to 28 percent, with clay mineral activity ratios of 0.68 to 0.83 suggesting illite as the dominant clay mineral, consistent with derivation from the tuffaceous parent rock. Occasional thin sand lenses (1 to 3 inches thick) were observed within this stratum. SPT N-values range from 7 to 15 blows per foot, indicating firm to stiff consistency. The stratum exhibits moderate swell potential.

Layer 3 — Alluvial Sand and Silt (10-12 feet to 20-23 feet bgs): Gray to tan, medium dense to dense, silty sand to sandy silt (SM/ML). This water-bearing stratum contains the groundwater table at approximately 14 feet bgs. SPT N-values increase from approximately 17 blows per foot at the top of the stratum to 29 blows per foot near the bedrock contact, indicating increasing density with depth. The contact with the underlying bedrock is gradational over approximately 1 to 2 feet, with increasing rock fragments and weathered bedrock clasts near the base.

Layer 4 — Bedrock: Challis Volcanics (20-23 feet to boring termination): Gray-green to brown, fine-grained sandstone and coarse siltstone, tuffaceous, weakly to moderately cemented. The rock is thinly to medium bedded with bedding dipping approximately 5 to 8 degrees to the south. The upper 2 to 3 feet are moderately weathered with reduced core recovery; below this zone the rock grades to slightly weathered to fresh. Occasional clay-filled discontinuities (1 to 5 mm aperture) are present along bedding planes and cross-joints, spaced at approximately 2 to 5 foot intervals. Rock quality designation (RQD) values range from 45 to 78 percent, averaging approximately 62 percent, indicating fair to good rock quality. The bedrock contact was encountered at depths ranging from 20.5 feet (BH-PP2-03) to 23.0 feet (BH-PP2-02), with an average depth of

approximately 21.6 feet bgs.

3.3 Groundwater

Groundwater was encountered in all four borings within the alluvial sand and silt stratum (Layer 3). Static water levels measured 24 hours after drilling completion ranged from 13.5 to 14.2 feet bgs. The water table is expected to fluctuate seasonally by approximately 2 feet based on regional hydrogeologic data and proximity to Cottonwood Creek. The groundwater is unconfined and recharged primarily by infiltration from the creek and seasonal precipitation. Dewatering will be required for excavations extending below the water table, particularly for the pump bay which extends to 12 feet below grade. The relatively permeable sand and silt stratum can be effectively dewatered using conventional sump pumping methods.

4.0 LABORATORY TESTING

Representative soil and rock samples were tested in the laboratory to characterize their engineering properties. Soil index and corrosion testing was performed by Intermountain Analytical Laboratories (IAL) of Grand Junction, Colorado, an AASHTO R18 accredited testing facility. Rock strength testing was performed in WGC's in-house rock mechanics laboratory in Boise, Idaho. All testing was performed in accordance with the applicable ASTM and AASHTO standards as noted.

4.1 Index Properties

Natural moisture content determinations (ASTM D2216) were performed on samples from each stratum. Results are summarized in the following table. Moisture contents in the clay stratum range from 17.1 to 22.1 percent, generally increasing with depth as samples approach the water table. These values are slightly above the plastic limit (average PL = 19 percent), indicating the clay is near or at full saturation in the lower portion of the stratum.

Table 4-1: Natural Moisture Content Summary

Sample ID	Boring	Depth (ft bgs)	Material	Moisture Content (%)
PP2-01-SS1	BH-PP2-01	2.0-3.5	Silty sand fill	8.2
PP2-01-SS2	BH-PP2-01	4.0-5.5	CL — lean clay	18.4
PP2-01-SS3	BH-PP2-01	7.0-8.5	CL — lean clay	21.2
PP2-01-SS4	BH-PP2-01	12.0-13.5	SM — silty sand	12.8
PP2-01-SS5	BH-PP2-01	15.0-16.5	SM — silty sand (sat)	16.5
PP2-02-SS2	BH-PP2-02	5.0-6.5	CL — lean clay	17.9
PP2-02-SS3	BH-PP2-02	8.0-9.5	CL — lean clay	22.1
PP2-02-SS4	BH-PP2-02	13.0-14.5	SM — silty sand	14.2
PP2-03-SS1	BH-PP2-03	2.0-3.0	Silty sand fill	7.5
PP2-03-SS2	BH-PP2-03	3.5-5.0	CL — lean clay	17.1
PP2-03-SS3	BH-PP2-03	7.5-9.0	CL — lean clay	20.8
PP2-04-SS2	BH-PP2-04	4.5-6.0	CL — lean clay	18.8
PP2-04-SS3	BH-PP2-04	8.5-10.0	CL — lean clay	21.5

4.2 Gradation and Hydrometer Analyses

Grain size distribution analyses were performed on four samples of the clay stratum (Layer 2) in accordance with ASTM D6913 (sieve analysis) and ASTM D7928 (hydrometer analysis). Results confirm that the clay stratum consists predominantly of fine-grained material with 55 to 64 percent passing the No. 200 sieve. The clay fraction (finer than 0.002 mm) ranges from 22.5 to 26.8 percent. A minor gravel component (0.5 to 2.8 percent) is present, consisting of subangular to subrounded volcanic rock fragments.

Table 4-2: Grain Size Distribution Summary — Layer 2 (CL)

Sample ID	Boring	Depth (ft bgs)	Gravel (%)	Sand (%)	Silt (%)	Clay (%)	D60 (mm)
PP2-01-SS2	BH-PP2-01	4.0-5.5	2.1	39.9	34.8	23.2	0.105

Sample ID	Boring	Depth (ft bgs)	Gravel (%)	Sand (%)	Silt (%)	Clay (%)	D60 (mm)
PP2-01-SS3	BH-PP2-01	7.0-8.5	0.8	37.2	36.0	26.0	0.078
PP2-02-SS3	BH-PP2-02	8.0-9.5	0.5	35.5	37.2	26.8	0.068
PP2-03-SS2	BH-PP2-03	3.5-5.0	2.8	42.2	32.5	22.5	0.120

4.3 Atterberg Limits

Atterberg limits (ASTM D4318) were determined on seven samples of the clay stratum. All samples classify as CL (lean clay) per the Unified Soil Classification System (ASTM D2487). Liquid limits range from 35 to 41, plastic limits from 18 to 21, and plasticity indices from 17 to 20. The samples plot below and parallel to the A-line on the plasticity chart, consistent with illitic clay mineralogy. Activity ratios (PI divided by clay fraction) range from 0.68 to 0.83, confirming illite as the dominant clay mineral. This is consistent with the tuffaceous origin of the parent rock, as the alteration of volcanic glass and feldspar in tuffaceous sediments typically produces illite and mixed-layer clay minerals.

Table 4-3: Atterberg Limits Summary — Layer 2 (CL)

Sample ID	Boring	Depth (ft bgs)	LL	PL	PI	USCS	Fines (%)	Clay (%)	Activity
PP2-01-SS2	BH-PP2-01	4.0-5.5	36	18	18	CL	58	23	0.78
PP2-01-SS3	BH-PP2-01	7.0-8.5	39	20	19	CL	62	26	0.73
PP2-02-SS2	BH-PP2-02	5.0-6.5	37	19	18	CL	57	24	0.75
PP2-02-SS3	BH-PP2-02	8.0-9.5	41	21	20	CL	64	27	0.74
PP2-03-SS2	BH-PP2-03	3.5-5.0	35	18	17	CL	55	23	0.74
PP2-03-SS3	BH-PP2-03	7.5-9.0	38	19	19	CL	61	26	0.73
PP2-04-SS2	BH-PP2-04	4.5-6.0	37	19	18	CL	59	24	0.75

4.4 Moisture-Density Relationships

Standard Proctor compaction tests (ASTM D698) were performed on two samples of the clay stratum to establish compaction criteria for structural fill placement. The maximum dry density ranged from 112.5 to 114.2 pounds per cubic foot (pcf) at optimum moisture contents of 15.9 to 16.8 percent.

Table 4-4: Standard Proctor Compaction Results

Sample ID	Boring	Depth (ft bgs)	Max Dry Density (pcf)	Optimum MC (%)
PP2-01-SS3	BH-PP2-01	7.0-8.5	112.5	16.8
PP2-03-SS2	BH-PP2-03	3.5-5.0	114.2	15.9

4.5 Swell and Consolidation

One-dimensional swell and consolidation tests (ASTM D4546) were performed on two relatively undisturbed Shelby tube samples of the clay stratum. The samples were tested under a seating load of 100 psf and exhibited free swell ranging from 2.52 to 3.10 percent, indicating moderate swell potential per the U.S. Bureau of Reclamation classification. Swell pressures ranged from 2,600 to 3,100 psf.

Consolidation test results indicate that the clay is overconsolidated with preconsolidation pressures of 2,500 to 3,000 psf, corresponding to overconsolidation ratios (OCR) of approximately 4 to 5 at the tested depths. This overconsolidation is attributed to desiccation and possibly ancient terrace loading. The compression index (C_c) ranges from 0.18 to 0.21 and the recompression index (C_r) from 0.035 to 0.038, which are typical values for lean clays of moderate plasticity.

Table 4-5: Swell and Consolidation Test Results

Sample ID	Boring	Depth (ft bgs)	Swell (%)	Swell Press. (psf)	Pc (psf)	C_c	C_r	e_0
PP2-01-ST1	BH-PP2-01	6.0-8.0	2.52	2,600	2,500	0.18	0.035	0.62
PP2-03-ST1	BH-PP2-03	5.5-7.5	3.10	3,100	3,000	0.21	0.038	0.65

4.6 Corrosion Testing

Corrosion testing was performed on three soil samples to evaluate the aggressiveness of the subsurface environment to buried concrete and steel. Testing included soil resistivity (ASTM G57/AASHTO T-288), pH (ASTM D4972), sulfate content (ASTM D4327), and chloride content (ASTM D4327). Results indicate that the clay stratum (Layer 2) is classified as "corrosive" to steel based on resistivity values below 2,000 ohm-cm. Sulfate concentrations of 780 to 890 ppm place the site in ACI 318 Sulfate Exposure Class S1, requiring the use of Type II portland cement for below-grade concrete. The sand stratum (Layer 3) is non-corrosive with resistivity exceeding 5,000 ohm-cm.

Table 4-6: Corrosion Test Results

Sample ID	Boring	Depth (ft bgs)	Material	Resist. (ohm-cm)	pH	Sulfate (ppm)	Chloride (ppm)
PP2-01-SS2	BH-PP2-01	4.0-5.5	CL clay	1,720	7.5	890	160
PP2-01-SS4	BH-PP2-01	12.0-13.5	SM sand	5,480	7.8	310	70
PP2-02-SS3	BH-PP2-02	8.0-9.5	CL clay	1,950	7.7	780	135

4.7 Unconfined Compressive Strength

Twelve unconfined compressive strength (UCS) tests were performed on rock core specimens in accordance with ASTM D7012 Method C. Specimens were prepared as right cylinders with a nominal diameter of 1.85 inches (NQ core) and a length-to-diameter ratio of approximately 2.1. All specimens were tested at their natural moisture content. The specimens were loaded axially at a constant stress rate until failure, with axial and (where instrumented) circumferential strain measured throughout the test.

Unconfined compressive strength values range from 5,180 to 8,680 psi with a mean of 7,166 psi and standard deviation of 1,053 psi. The secant modulus at 50 percent of peak strength (E50) ranges from 1,180 to 1,810 ksi with a mean of 1,510 ksi. Poisson's ratio values range from 0.20 to 0.24 with a mean of 0.22. The two lowest strength values (5,180 and 6,250 psi) correspond to specimens containing visible clay-filled discontinuities, which controlled the failure mode. Excluding these specimens, the mean UCS is 7,557 psi. The rock is classified as "moderately weak" to "medium strong" per the ISRM classification (1 to 25 MPa = 145 to 3,625 psi is weak; 25 to 50 MPa = 3,625 to 7,250 psi is medium strong).

Table 4-7: Unconfined Compressive Strength Results (* = clay-filled discontinuity controlled failure)

Specimen ID	Boring	Depth (ft bgs)	D (in)	L (in)	UCS (psi)	E50 (ksi)	ν	Failure Mode
PP2-01-RC1-U1	BH-PP2-01	22.0-22.5	1.85	3.92	6,840	1,420	0.21	Axial split
PP2-01-RC1-U2	BH-PP2-01	23.5-24.0	1.85	3.88	7,520	1,580	0.23	Shear 60°
PP2-01-RC1-U3	BH-PP2-01	25.0-25.5	1.85	3.95	5,180*	1,180	0.20	Axial (clay disc)
PP2-01-RC2-U1	BH-PP2-01	30.0-30.5	1.85	3.90	8,240	1,720	0.22	Cone/shear
PP2-01-RC2-U2	BH-PP2-01	33.0-33.5	1.85	3.94	8,680	1,810	0.24	Shear 55°
PP2-02-RC1-U1	BH-PP2-02	24.0-24.5	1.85	3.91	7,180	1,490	0.22	Axial split
PP2-02-RC1-U2	BH-PP2-02	26.0-26.5	1.85	3.87	6,250*	1,350	0.21	Shear (clay disc)
PP2-02-RC1-U3	BH-PP2-02	28.5-29.0	1.85	3.93	8,050	1,680	0.23	Cone/shear
PP2-03-RC1-U1	BH-PP2-03	21.5-22.0	1.85	3.89	5,890	1,250	0.20	Axial split
PP2-03-RC1-U2	BH-PP2-03	23.0-23.5	1.85	3.96	7,740	1,610	0.22	Shear 65°
PP2-04-RC1-U1	BH-PP2-04	22.5-23.0	1.85	3.90	6,520	1,380	0.21	Axial split
PP2-04-RC1-U2	BH-PP2-04	24.0-24.5	1.85	3.92	7,900	1,650	0.23	Shear 55°

* Specimen contained clay-filled discontinuity that controlled the failure surface. These values represent lower-bound strength estimates for the rock mass.

4.8 Indirect Tensile Strength

Fourteen indirect tensile strength (ITS) tests were performed on rock core disc specimens in accordance with ASTM D3967. Specimens were prepared with a nominal diameter of 1.85 inches and thickness of approximately 0.93 inches (thickness-to-diameter ratio of approximately 0.5). Loading was applied diametrically through curved loading jaws at a constant rate until tensile splitting failure occurred.

Indirect tensile strength values range from 317 to 514 psi with a mean of 438 psi and standard deviation of 63 psi. Peak loads range from 840 to 1,420 lbf. The ratio of mean UCS to mean ITS is 16.4, which is within the expected range of 8 to 20 for sedimentary rocks. As with the UCS tests, the lowest ITS values (317 and 325 psi) correspond to specimens intersected by clay-filled discontinuities, which induced off-center or premature failure.

Table 4-8: Indirect Tensile Strength Results (* = clay-filled discontinuity present)

Specimen ID	Boring	Depth (ft bgs)	D (in)	t (in)	Peak (lbf)	ITS (psi)	Failure
PP2-01-RC1-T1	BH-PP2-01	22.5-23.0	1.85	0.94	1,280	468	Central
PP2-01-RC1-T2	BH-PP2-01	24.0-24.3	1.85	0.96	1,190	426	Central
PP2-01-RC1-T3	BH-PP2-01	24.5-24.8	1.85	0.91	840*	317	Off-center
PP2-01-RC2-T1	BH-PP2-01	30.5-30.8	1.85	0.95	1,420	514	Central
PP2-01-RC2-T2	BH-PP2-01	31.0-31.3	1.85	0.93	1,380	510	Central
PP2-02-RC1-T1	BH-PP2-02	24.5-24.8	1.85	0.94	1,310	479	Central
PP2-02-RC1-T2	BH-PP2-02	25.0-25.3	1.85	0.92	1,150	430	Central
PP2-02-RC1-T3	BH-PP2-02	27.0-27.3	1.85	0.95	1,350	489	Central
PP2-03-RC1-T1	BH-PP2-03	22.0-22.3	1.85	0.93	980	362	Central
PP2-03-RC1-T2	BH-PP2-03	22.5-22.8	1.85	0.94	1,220	446	Central
PP2-03-RC1-T3	BH-PP2-03	23.5-23.8	1.85	0.91	1,100	415	Off-center
PP2-04-RC1-T1	BH-PP2-04	23.0-23.3	1.85	0.95	1,260	456	Central
PP2-04-RC1-T2	BH-PP2-04	23.5-23.8	1.85	0.92	870*	325	Central (clay)
PP2-04-RC1-T3	BH-PP2-04	24.0-24.3	1.85	0.94	1,340	490	Central

4.9 Core Recovery and RQD

Core recovery and Rock Quality Designation (RQD) were determined for each core run in accordance with ASTM D6032. Core recovery ranges from 84 to 95 percent and RQD ranges from 45 to 78 percent. Both parameters generally improve with depth as the degree of weathering decreases. The upper 2 to 3 feet of bedrock consistently shows the lowest recovery and RQD values, attributed to increased fracturing and weathering at the alluvium-bedrock interface.

Table 4-9: Core Recovery and RQD Summary

Core Run	Boring	Interval (ft bgs)	Length (ft)	Recovery (%)	RQD (%)	Notes
PP2-01-RC1	BH-PP2-01	21.0-26.0	5.0	88	62	Clay disc at 25.2 ft
PP2-01-RC2	BH-PP2-01	26.0-35.0	9.0	92	71	Improved quality below 28 ft
PP2-01-RC3	BH-PP2-01	35.0-40.0	5.0	95	78	Competent, fewer joints
PP2-02-RC1	BH-PP2-02	23.0-30.0	7.0	86	58	2 clay-filled discs
PP2-02-RC2	BH-PP2-02	30.0-35.0	5.0	91	68	Improving with depth
PP2-03-RC1	BH-PP2-03	20.5-30.0	9.5	84	55	Most weathered (near creek)
PP2-04-RC1	BH-PP2-04	22.0-25.0	3.0	87	60	Short run, typical quality

Standard penetration test results from all four borings are summarized in the following table. N-values show a consistent increasing trend with depth across all boring locations, with no anomalous soft zones or voids encountered. Refusal on bedrock was reached at depths ranging from 20.5 to 23.0 feet bgs.

Table 4-10: Standard Penetration Test Results (N-values, blows/ft)

Depth (ft bgs)	BH-PP2-01	BH-PP2-02	BH-PP2-03	BH-PP2-04	Layer
2.5	10	8	11	9	Fill / topsoil
5.0	7	9	8	7	CL — lean clay
7.5	11	10	12	10	CL — lean clay
10.0	13	14	11	15	CL / SM transition
12.5	17	16	19	15	SM — silty sand
15.0	21	20	23	19	SM — silty sand (sat)
17.5	24	26	22	25	SM — silty sand
20.0	27	29	28	26	SM / bedrock transition
22.0	32/6"	35/5"	30	33/6"	Bedrock (refusal)
25.0	50/4"	50/3"	50/5"	50/4"	Bedrock

5.0 ENGINEERING ANALYSIS

5.1 Foundation Alternatives

Two foundation alternatives were evaluated for the pump bay structure: (1) spread footings bearing on the alluvial clay stratum (Layer 2), and (2) drilled shafts extending to bedrock (Layer 4). The pump bay extends approximately 12 feet below existing grade, which places the footing base within the clay stratum at approximately 14 to 16 feet below original grade (i.e., within the lower portion of Layer 2 or upper portion of Layer 3). Given the swell potential of the clay, spread footings require swell mitigation measures as described in Section 6.2.

For the heavier equipment pads (vertical turbine pump foundations with significant dynamic loads), drilled shafts socketed into bedrock are recommended to minimize settlement and provide positive anchorage against uplift and overturning moments. The discharge manifold and motor control center may be supported on spread footings on the clay stratum, provided swell mitigation is implemented.

5.2 Bearing Capacity

Allowable bearing capacity was evaluated using Terzaghi's bearing capacity equation with appropriate modifications for footing shape and depth. For spread footings on the clay stratum (Layer 2), the allowable net bearing pressure is 3,000 psf based on a factor of safety of 3.0 against general shear failure. This assumes footings with a minimum width of 3 feet, founded at a minimum depth of 4 feet below finished grade (below the zone of seasonal moisture fluctuation and swell activity).

For drilled shafts bearing on or socketed into bedrock (Layer 4), the allowable end-bearing pressure is 20,000 psf based on the lower-bound UCS value of 5,180 psi and a factor of safety of 10 (conservative, accounting for potential clay-filled discontinuities in the bearing zone). Side friction in the rock socket may be taken as 15 percent of the mean UCS (approximately 1,075 psi), with an allowable value of 215 psi (factor of safety of 5). Drilled shafts should extend a minimum of 5 feet (approximately 3 shaft diameters for a 20-inch shaft) into competent bedrock below the weathered zone.

5.3 Settlement

Settlement under the proposed foundation loads was estimated using the consolidation test results and elastic theory. For spread footings on the clay stratum loaded at the allowable bearing pressure of 3,000 psf, the estimated total settlement is 0.75 to 1.25 inches. Approximately 80 percent of this settlement is expected to occur as immediate (elastic) settlement, with the remaining 20 percent occurring as primary consolidation over a period of 6 to 12 months. The overconsolidated nature of the clay (OCR of 4 to 5) limits consolidation settlement because foundation loads are expected to remain below the preconsolidation pressure. Differential settlement over a horizontal distance of 50 feet is estimated at less than 0.5 inches, which is within acceptable limits for the proposed structure.

For drilled shaft foundations bearing on bedrock, settlement is expected to be negligible (less than 0.25 inches) given the high modulus of the rock mass.

5.4 Lateral Earth Pressures

Lateral earth pressure parameters for the retaining wall along the access road (BH-PP2-04 location) were computed using Rankine earth pressure theory with the following parameters for the alluvial clay: friction angle = 28 degrees, cohesion = 200 psf, unit weight = 120 pcf. The resulting coefficients and equivalent fluid

pressures are:

Active earth pressure coefficient $K_a = 0.35$ (equivalent fluid weight 40 pcf); at-rest earth pressure coefficient $K_0 = 0.50$ (equivalent fluid weight 58 pcf); passive earth pressure coefficient $K_p = 2.9$. The retaining wall should be designed for at-rest pressures if the wall is rigid or restrained at the top (as in the case of a basement wall). A surcharge of 250 psf should be added for traffic and construction equipment loads behind the wall.

5.5 Seismic Considerations

The site is classified as Site Class D (stiff soil) per ASCE 7-16 based on the average shear wave velocity of the upper 100 feet, estimated from SPT correlations. The mapped seismic hazard parameters for the site (per USGS, 2014) are: peak ground acceleration $PGA = 0.15g$, short-period spectral acceleration $S_s = 0.35g$, and 1-second spectral acceleration $S_1 = 0.13g$, all for the Maximum Considered Earthquake (MCE) with 2 percent probability of exceedance in 50 years.

A liquefaction screening analysis was performed for the sand and silt stratum (Layer 3) using the simplified procedure of Youd et al. (2001). The analysis indicates that liquefaction is not a concern at this site because: (a) the overlying clay stratum provides a confining cap; (b) SPT N-values in the sand exceed the critical values for the design earthquake; and (c) the fines content of the sand stratum (typically 15 to 25 percent) increases the cyclic resistance ratio. The factor of safety against liquefaction exceeds 1.5 at all depths.

6.0 RECOMMENDATIONS

6.1 Foundation Design

Based on the results of our investigation and analysis, we recommend the following foundation system for the Cottonwood Creek Pumping Plant (PP-2):

Pump Bay and Equipment Pads: Drilled shafts socketed a minimum of 5 feet into competent bedrock (below the weathered zone). Shaft diameter should be sized based on structural loads; a minimum diameter of 18 inches is recommended to facilitate clean-out inspection. Allowable end-bearing: 20,000 psf. Allowable side friction in rock socket: 215 psi. Shafts should be tremie-grouted if groundwater is encountered during installation.

Discharge Manifold and Motor Control Center: Spread footings bearing on the alluvial clay stratum (Layer 2) at a minimum depth of 4 feet below finished grade. Allowable net bearing pressure: 3,000 psf. Minimum footing width: 3 feet. All footings should bear on a minimum 24-inch layer of compacted structural fill placed on properly prepared subgrade (see Section 6.2 for swell mitigation requirements).

Access Road Retaining Wall: Continuous spread footing bearing on the clay stratum at a minimum depth of 3 feet below finished grade. Design for at-rest earth pressures (equivalent fluid weight 58 pcf) plus 250 psf traffic surcharge. Provide drainage behind the wall using free-draining granular backfill and a perforated drain pipe at the footing level.

6.2 Swell Mitigation

The clay stratum exhibits moderate swell potential (2.5 to 3.1 percent free swell). To mitigate the risk of swell-induced distress to the structure, we recommend the following:

Over-Excavation and Replacement: Excavate a minimum of 2 feet below the bottom of footings and replace with compacted non-expansive structural fill. The replacement fill should be a well-graded granular material (USCS classification GW, GP, SW, or SP) with a maximum particle size of 3 inches, compacted to a minimum of 95 percent of the maximum dry density per ASTM D698. The fill should be placed at a moisture content within 2 percent of optimum.

Moisture Barriers: Install a polyethylene moisture barrier (minimum 10-mil thickness) on all sides of the foundation excavation to limit moisture migration into the clay subgrade. Extend the barrier a minimum of 5 feet beyond the building footprint. Grade the site to direct surface water away from the foundation with a minimum slope of 5 percent within 10 feet of the structure.

Void Space: For grade beams spanning between drilled shafts, provide a minimum 4-inch void space beneath the beam to allow for potential upward swell of the clay. The void may be created using cardboard void forms that deteriorate after placement, allowing the clay to swell without loading the structure.

6.3 Corrosion Protection

Based on the corrosion test results, the following protective measures are recommended for all below-grade concrete and steel:

Concrete: Use Type II portland cement conforming to ASTM C150 for all below-grade concrete. The maximum water-cement ratio should not exceed 0.45, and the minimum 28-day compressive strength should be 4,500 psi. A minimum concrete cover of 3 inches is recommended for reinforcing steel in contact with soil.

Steel: All embedded steel (anchor bolts, H-piles, sheet piling) should be protected with a fusion-bonded epoxy coating conforming to ASTM A775 or hot-dip galvanizing conforming to ASTM A123. Alternatively, a corrosion allowance of 1/16-inch additional thickness may be provided for uncoated steel in contact with the clay stratum.

6.4 Excavation and Dewatering

Excavation through the alluvial overburden (Layers 1 through 3) can be accomplished with conventional earthmoving equipment. The clay stratum (Layer 2) will stand on temporary cut slopes of 1H:1V (horizontal to vertical) to a maximum height of 8 feet, provided the excavation remains open for less than 30 days and no surcharge loads are placed within 5 feet of the excavation edge. For deeper excavations extending into the sand stratum (Layer 3), flatter slopes of 1.5H:1V are recommended due to the granular nature of the material and potential for groundwater seepage.

Dewatering will be required for the pump bay excavation, which extends approximately 12 feet below grade and will encounter the water table at approximately 14 feet bgs. Given the moderate permeability of the sand stratum, conventional sump pumping from the excavation floor is expected to be adequate. An estimated pumping rate of 30 to 60 gallons per minute should be anticipated during excavation and forming operations. A contingency for wellpoint dewatering should be included in the construction budget in the event that sump pumping proves inadequate during peak groundwater conditions.

6.5 Construction Monitoring

We recommend that WGC be retained to provide geotechnical construction monitoring services during the following critical phases of construction: (a) foundation excavation observation and bearing surface verification for spread footings; (b) drilled shaft installation inspection including socket depth verification, cleanliness inspection, and concrete placement observation; (c) structural fill placement inspection and compaction testing; and (d) retaining wall backfill and drainage installation inspection. These services are essential to verify that the subsurface conditions encountered during construction are consistent with those observed during the investigation and that the geotechnical recommendations are properly implemented.

7.0 LIMITATIONS

This report has been prepared for the exclusive use of the Clearwater Valley Irrigation District for the specific project described herein. The recommendations contained in this report are based on the subsurface conditions encountered at the specific boring locations at the time of our exploration. Subsurface conditions may vary between boring locations and may change over time due to natural processes or human activities.

The borings were advanced at specific locations and to specific depths. They reveal subsurface conditions only at those specific locations and only to the depths explored. Conditions between borings are inferred based on geologic interpretation and professional judgment. If subsurface conditions different from those described in this report are encountered during construction, WGC should be notified immediately so that the recommendations can be re-evaluated.

The analyses and recommendations in this report are based on the project information available at the time of report preparation. If the project design is modified (including changes to building location, size, configuration, or loading), the recommendations in this report should be reviewed and may require revision.

This report should not be used by other parties or for other projects without written authorization from WGC. Any party who relies on this report without such authorization does so at their own risk. WGC's liability is limited to the fee paid for the services described herein.

The findings, opinions, and recommendations presented in this report are based on our understanding of current geotechnical engineering practice in the region and are made in accordance with generally accepted geotechnical engineering principles. No warranty, expressed or implied, is made.